Heat-Flow Transport of Turán Discriminants for Riemann's Ξ-Function: the Hyperbolic Entropic Flow, Corrected

Biswajit Mondal
Developer and Researcher in Mathematical Physics

June 9, 2026; revised October 16, 2026

Abstract

We deform the Taylor coefficients of Riemann's Ξ-function under the de Bruijn–Newman heat flow. We follow the evolution of the degree-two Jensen (Turán) discriminants of the factorial-normalized coefficients. The main result is an exact transport identity: $\partial_t \log(\bar{\Delta}(n,t)+1) = -\Delta^2\rho_n(t)$. Here $\bar{\Delta}$ is the normalized Turán discriminant and $\rho_n = M_{n+1}/M_n$ is the ratio of consecutive kernel moments. We also record a normalization trap. For the raw kernel moments the quadratic discriminant is negative at every index and every time, by the Cauchy–Schwarz inequality. A weighted logarithmic "entropy" of the Turán discriminants is well defined when they are positive, which at $t = 0$ is a classical theorem, and when they decay at most polynomially. Its derivative follows from the transport identity. Degree two alone cannot reach the Riemann Hypothesis (RH), and we state precisely what additional input would be needed.

Correction notice. This version replaces the June 2026 draft. That draft built its Jensen polynomials from raw kernel moments b_m, treating them as if they were Taylor coefficients of Ξ. For those moments the quadratic discriminant is negative for every n and t (Proposition 1), so its entropy functional was undefined. The draft also substituted $b_{n+1}/b_n = -\rho_n$ although $b_m > 0$, which reversed the sign of its transport law. It further asserted that monotonicity of the functional is equivalent to RH, and that "Dyson stiffness" of GUE spacings protects that monotonicity. Both assertions are withdrawn. The first fails logically (Section 6); the second is a statistical heuristic, not an argument. The draft's claim that the GORZ threshold grows like e^{Cd} was unsupported and has been removed.

1 Introduction

RH is equivalent to the hyperbolicity (all roots real) of every Jensen polynomial attached to the factorial-normalized Taylor coefficients of Ξ [3, 1]. Griffin, Ono, Rolen and Zagier proved that for each fixed degree d these polynomials are hyperbolic for all sufficiently large shifts n [8]. Degree two, the Turán inequalities, holds for every shift [6]. Degree three is due to Dimitrov and Lucas [7].

We study how these discriminants move under the de Bruijn–Newman heat flow [4, 5]. Increasing the flow time makes zeros real; RH is the statement that the de Bruijn–Newman constant satisfies $\Lambda \le 0$. It is known that $0 \le \Lambda \le 0.2$ [10, 9, 11]. Our contribution is modest and exact: a closed transport law for the order-two discriminants, a correct normalization, and a sharp statement of what such a law can and cannot give.

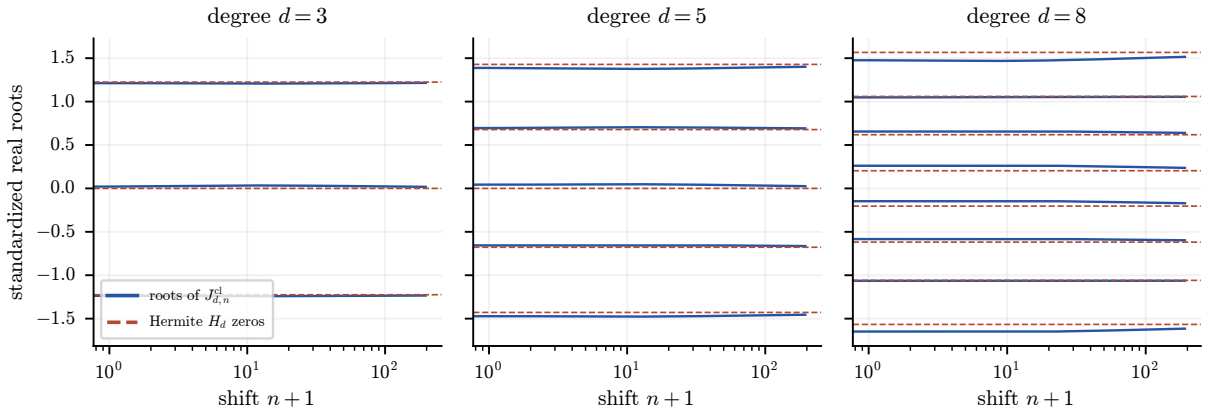


Figure 1: The Hermite limit behind the GORZ theorem, computed from midpoints of rigorous enclosures of the first 201 Taylor coefficients of Ξ. For degrees $d = 3, 5, 8$ and every available shift n, all roots of the classical (factorial-normalized) Jensen polynomial $J^{\mathrm{cl}}_{d,n}(X) = \sum_j \binom{d}{j}\gamma_{n+j}X^j$, $\gamma_m = m!\,\mu_m$, $\mu_m = (-1)^m[z^{2m}]\Xi(z)$, are real. After centring and scaling, they lie close to the zeros of H_d (dashed). The figure covers $n \le 200 - d$ only; it is an illustration of the asymptotic regime, not a certificate.

2 The heat flow and its coefficients

We use de Bruijn's normalization, in which Λ is defined. Let

$$\Phi(u) = \sum_{k\ge1}\bigl(2\pi^2k^4e^{9u} - 3\pi k^2e^{5u}\bigr)\exp\bigl(-\pi k^2e^{4u}\bigr), \qquad u \ge 0, \tag{1}$$

and

$$H_t(z) = \int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,du, \qquad H_0(z) = \tfrac{1}{8}\,\Xi(z/2). \tag{2}$$

The kernel is positive [2] and decays double-exponentially, so every moment

$$M_m(t) = \int_0^\infty e^{tu^2}\Phi(u)\,u^{2m}\,du > 0 \tag{3}$$

is finite for every real t. Expanding the cosine,

$$H_t(z) = \sum_{m\ge0}\frac{(-1)^m M_m(t)}{(2m)!}z^{2m} = F_t(-z^2), \qquad F_t(x) = \sum_{m\ge0}\mu_m(t)x^m, \quad \mu_m = \frac{M_m}{(2m)!} > 0.$$

Put $\gamma_m = m!\,\mu_m$ and define the Jensen polynomials $J_t^{d,n}(X) = \sum_{j=0}^d \binom{d}{j}\gamma_{n+j}(t)X^j$. All zeros of H_t are real exactly when all zeros of F_t are real (and then negative, since $\mu_m > 0$). By the Jensen–Pólya theory [3, 1], this holds exactly when every $J_t^{d,n}$ is hyperbolic. By the definition of Λ, it holds exactly when $t \ge \Lambda$.

Lemma 1 (Moment flow)**.** *$\partial_t M_m(t) = M_{m+1}(t)$ for all $m \ge 0$ and $t \in \mathbb{R}$.*

Proof. Differentiate under the integral; double-exponential decay of Φ justifies dominated convergence locally uniformly in t. □

Lemma 2 (Coefficient flow)**.** *$\partial_t\gamma_m = 2(2m+1)\,\gamma_{m+1}$.*

Proof. By Lemma 1, $\partial_t\gamma_m = m!\,M_{m+1}/(2m)!$, while $\gamma_{m+1} = (m+1)!\,M_{m+1}/(2m+2)!$. Their ratio is $(2m+2)(2m+1)/(m+1) = 2(2m+1)$. □

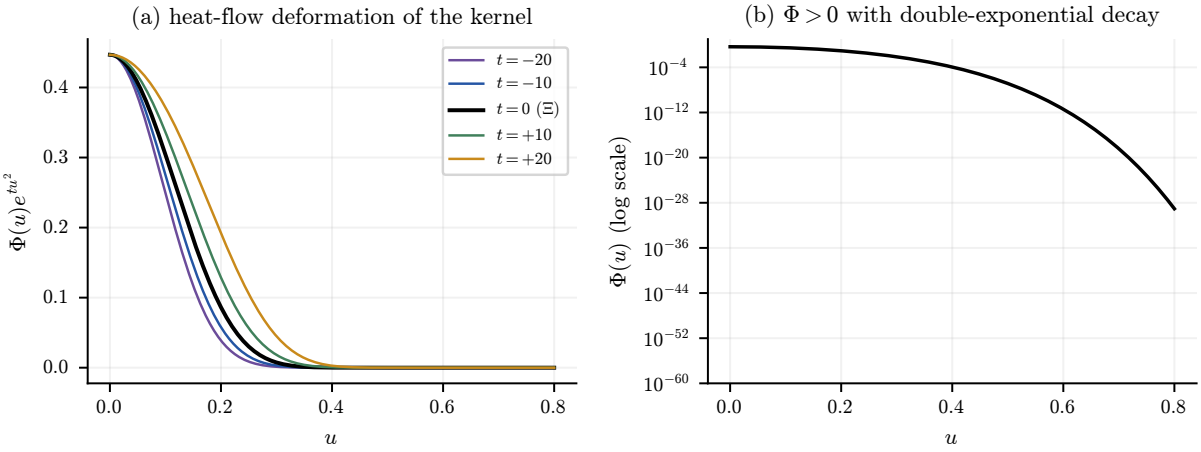


Figure 2: The deformation in kernel form, computed from the theta series for Φ. (a) The deformed kernels $\Phi(u)e^{tu^2}$; the values of t are enlarged for visibility. Positive t moves weight to larger u and so raises the higher moments, as $\partial_t M_m = M_{m+1}$ records. (b) Φ is positive and decays double-exponentially, which is what makes every moment finite for every t.

3 Two discriminants, one of them always negative

Proposition 1 (The raw-moment trap)**.** *For every $n \geq 0$ and $t \in \mathbb{R}$, $M_{n+1}(t)^2 < M_n(t)M_{n+2}(t)$. Consequently the quadratic $M_n + 2M_{n+1}X + M_{n+2}X^2$ has two nonreal roots for every n and t.*

Proof. Apply the Cauchy–Schwarz inequality to u^{2n} and u^{2n+4} in $L^2(e^{tu^2}\Phi(u)\,du)$. Equality would force u^2 to be constant on the support of Φ, which is $[0,\infty)$. □

The correct object uses the factorial normalization. Define the *normalized Turán discriminant*

$$\bar{\Delta}(n,t) = \frac{\gamma_{n+1}(t)^2}{\gamma_n(t)\gamma_{n+2}(t)} - 1, \qquad \operatorname{disc} J_t^{2,n} = 4\gamma_n\gamma_{n+2}\,\bar{\Delta}(n,t). \tag{4}$$

Thus $J_t^{2,n}$ is hyperbolic if and only if $\bar{\Delta}(n,t) \geq 0$. A direct computation gives

$$\bar{\Delta}(n,t) + 1 = \frac{2n+3}{2n+1} \cdot \frac{M_{n+1}^2}{M_n M_{n+2}}, \tag{5}$$

so by Proposition 1 the Turán condition is the band $\frac{2n+1}{2n+3} \leq M_{n+1}^2/(M_nM_{n+2}) < 1$, of width $2/(2n+3)$. In particular $-1 < \bar{\Delta}(n,t) < \frac{2}{2n+1} \leq 2$. Two facts are known:

- $\bar{\Delta}(n,0) > 0$ for every $n \geq 0$. These are the Turán inequalities for Ξ, proved by Csordas, Norfolk and Varga [6].
- $\bar{\Delta}(n,t) \geq 0$ for every n when $t \geq \Lambda$, since then F_t lies in the Laguerre–Pólya class.

4 The transport identity

Theorem 1 (Turán transport)**.** *Let $\rho_n(t) = M_{n+1}(t)/M_n(t)$ and $\Delta^2\rho_n = \rho_{n+2} - 2\rho_{n+1} + \rho_n$. For all $n \geq 0$ and $t \in \mathbb{R}$,*

$$\frac{\partial}{\partial t}\log\bigl(\bar{\Delta}(n,t)+1\bigr) = -\Delta^2\rho_n(t), \qquad \textit{equivalently} \qquad \frac{\partial\bar{\Delta}}{\partial t} = -(\bar{\Delta}+1)\,\Delta^2\rho_n. \tag{6}$$

Proof. By Lemma 1, $\partial_t\rho_n = (M_{n+2}M_n - M_{n+1}^2)/M_n^2 = \rho_n(\rho_{n+1} - \rho_n)$, so $\partial_t \log\rho_n = \rho_{n+1} - \rho_n$. By (5), $\log(\bar{\Delta}+1) = \log\frac{2n+3}{2n+1} + \log\rho_n - \log\rho_{n+1}$, whose first term is independent of t. Differentiating, $(\rho_{n+1} - \rho_n) - (\rho_{n+2} - \rho_{n+1}) = -\Delta^2\rho_n$. □

Remark 1. The identity concerns the kernel moments only, so it holds for any positive kernel with finite moments, not just for Ξ. Its value is that it turns the time evolution of every Turán discriminant into a convexity statement about a single sequence. Central finite differences at step 10^{-8}, computed with 40-digit quadrature, agree with (6) to a relative error below 10^{-19} for $0 \le n \le 5$ at $t = 0$.

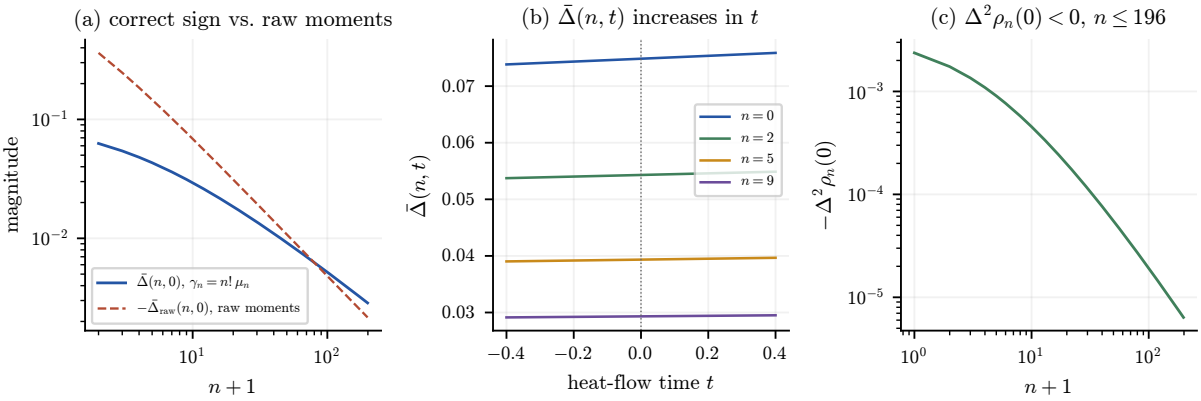


Figure 3: (a) The normalized Turán discriminant $\bar{\Delta}(n,0)$ of the factorial-normalized coefficients is positive and decays roughly like $0.57/n$. The raw-moment discriminant is negative for every n (Proposition 1); its absolute value is dashed. (b) $\bar{\Delta}(n,t)$ for $t \in [-0.4, 0.4]$, from quadrature of the moments: each increases with t. (c) $-\Delta^2\rho_n(0) > 0$ for $0 \le n \le 196$, the sign that makes (b) increasing through Theorem 1. Panels (a) and (c) use midpoints of certified coefficient balls and are not certificates. Turán ratios are invariant under the rescaling $z \mapsto z/2$, and the sign of $\Delta^2\rho_n$ is invariant too.

Observation. Using midpoints of certified enclosures of the first 201 coefficients, $\Delta^2\rho_n(0) < 0$ for every $0 \le n \le 196$. By Theorem 1, each of these Turán discriminants is increasing in t at $t = 0$.

Question 1. *Is $\Delta^2\rho_n(t) < 0$ for all $n \ge 0$ and all $t \ge 0$? Equivalently, is the moment-ratio sequence of $e^{tu^2}\Phi(u)$ strictly concave? An affirmative answer would make every Turán ratio nondecreasing along the forward flow. It concerns degree two only, so it would not bear on RH.*

5 The entropy functional, correctly defined

Definition 1. *Let $T^+ = \{t : \bar{\Delta}(n,t) > 0$ for all $n\}$. For $\alpha > 0$ and $t \in T^+$, put $\mathcal{H}_\alpha(t) = \sum_{n\ge 0} e^{-\alpha n} \log \bar{\Delta}(n,t)$.*

Proposition 2 (Convergence)**.** *Let $t \in T^+$, and suppose $\bar{\Delta}(n,t) \ge c\,(n+1)^{-K}$ for some $c, K > 0$ and all n. Then the series for $\mathcal{H}_\alpha(t)$ converges absolutely.*

Proof. From $\bar{\Delta} < 2$ and the hypothesis, $|\log \bar{\Delta}(n,t)| \le \log 2 + |\log c| + K \log(n+1)$. The series $\sum e^{-\alpha n} \log(n+1)$ converges. □

At $t = 0$ the positivity hypothesis is the Csordas–Norfolk–Varga theorem. For the polynomial lower bound we have only evidence: $n\,\bar{\Delta}(n,0)$ increases from 0.28 at $n = 10$ to 0.57 at $n = 197$. Effective forms of the GORZ asymptotics should prove a bound of the form $\bar{\Delta}(n,0) \ge c/(n+1)$; we have not carried this out. On any interval of T^+ where the lower bound

holds uniformly and the differentiated series converges locally uniformly, Theorem 1 gives

$$\frac{d\mathcal{H}_\alpha}{dt} = -\sum_{n\geq 0} e^{-\alpha n}\,\frac{\bar{\Delta}(n,t)+1}{\bar{\Delta}(n,t)}\,\Delta^2\rho_n(t). \tag{7}$$

6 What this framework can and cannot give

1. **Degree two is not RH-strength.** At $t = 0$ every Turán discriminant is already positive by [6]. A statement about $\mathcal{H}_\alpha$ alone therefore adds nothing toward RH, which needs all degrees d simultaneously.

2. **The direction of monotonicity.** If $d\mathcal{H}_\alpha/dt \geq 0$ on $[0,\infty)$, integration bounds $\mathcal{H}_\alpha(0)$ from *above* by later values. It cannot keep $\mathcal{H}_\alpha(0)$ away from $-\infty$. Transporting positivity from $t \geq \Lambda$, where it holds, back to $t = 0$ needs backward-in-time control. That is either $d\mathcal{H}/dt \leq 0$ on $[0,\Lambda]$, or a bound on $|d\mathcal{H}/dt|$ there. This is exactly as hard as the original problem unless it can be made uniform in d.

3. **The open input.** An RH-relevant version would need transport identities for the degree-d discriminants of $J_t^{d,n}$, together with backward control uniform in d. We do not know such identities for $d \geq 3$; finding them is the natural next step.

4. **Withdrawn heuristics.** Statistical properties of zeta zeros (pair correlation, GUE spacing) are averaged statements. They give no deterministic sign information about any individual $\Delta^2\rho_n(t)$, and they are not used here.

7 Conclusion

Corrected, the Hyperbolic Entropic Flow is a bookkeeping device for an exact identity: under the de Bruijn–Newman flow, the logarithm of each factorial-normalized Turán ratio moves at minus the second difference of the kernel moment ratios. The identity, its numerical confirmation and the raw-moment trap are solid. The observed concavity of moment ratios (Question 1) is a clean, kernel-only conjecture. Extending the transport law to higher-degree discriminants is the open problem that would connect this framework to RH. Nothing here proves RH or reduces it to a statement shown to be more tractable.

References

[1] J. L. W. V. Jensen, *Recherches sur la théorie des équations*, Acta Math. 36 (1913), 181–195.

[2] G. Pólya, *Bemerkung über die Integraldarstellung der Riemannschen ξ-Funktion*, Acta Math. 48 (1926), 305–317.

[3] G. Pólya, *Über die algebraisch-funktionentheoretischen Untersuchungen von J. L. W. V. Jensen*, Kgl. Danske Vid. Sel. Math.-Fys. Medd. 7 (1927), 3–33.

[4] N. G. de Bruijn, *The roots of trigonometric integrals*, Duke Math. J. 17 (1950), 197–226.

[5] C. M. Newman, *Fourier transforms with only real zeros*, Proc. Amer. Math. Soc. 61 (1976), 245–251.

[6] G. Csordas, T. S. Norfolk, R. S. Varga, *The Riemann hypothesis and the Turán inequalities*, Trans. Amer. Math. Soc. 296 (1986), 521–541.

[7] D. K. Dimitrov, F. R. Lucas, *Higher order Turán inequalities for the Riemann ξ-function*, Proc. Amer. Math. Soc. 139 (2011), 1013–1022.

[8] M. Griffin, K. Ono, L. Rolen, D. Zagier, *Jensen polynomials for the Riemann zeta function and other sequences*, Proc. Natl. Acad. Sci. USA 116 (2019), 11103–11110.

[9] D. H. J. Polymath, *Effective approximation of heat flow evolution of the Riemann ξ function, and a new upper bound for the de Bruijn–Newman constant*, Res. Math. Sci. 6 (2019), 31.

[10] B. Rodgers, T. Tao, *The de Bruijn–Newman constant is non-negative*, Forum Math. Pi 8 (2020), e6.

[11] D. Platt, T. Trudgian, *The Riemann hypothesis is true up to* $3 \cdot 10^{12}$, Bull. London Math. Soc. 53 (2021), 792–797.